

Unbiased estimation via randomization - Application to Option Pricing

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(Q1) 1.

$$\begin{aligned}
 \|Y_l - Y\|_2^2 &= E[Y_l - Y]^2 \\
 &= E[Y_l^2 - 2Y_l Y + Y^2] \\
 &= E[Y_l^2] - 2E[Y_l]E[Y] + E[Y^2] \quad \text{by independence of } Y_l \text{ and } Y \\
 &\rightarrow E[Y^2] - 2E[Y]^2 + E[Y^2] \\
 &= 2 \underbrace{(E[Y^2] - E[Y]^2)}_{\text{Var}[Y]} \\
 &= 2\text{Var}[Y] \neq 0 \quad \text{if } \text{Var}[Y] \neq 0
 \end{aligned}$$

By assumption, N is almost surely finite, i.e.

$$P(N = \infty) = 0.$$

But

$$\{N \geq 1\} \supset \{N \geq 2\} \supset \{N \geq 3\} \supset \dots$$

and $\cap_{l \geq 1} \{N \geq l\} = \{N = \infty\}$. so by continuity of probability,

$$0 = P(N = \infty) = \lim_{l \rightarrow \infty} P(N \geq l),$$

so for l big enough

$$\frac{\|Y_{l-1} - Y\|_2^2}{P(N \geq l)} \geq \frac{\text{Var}[Y]}{P(N \geq l)}.$$

But since $P(N \geq l) \rightarrow 0$ we have

$$\frac{\text{Var}[Y]}{P(N \geq l)} \rightarrow \infty$$

so

$$\sum_{l=1}^{\infty} \frac{\|Y_{l-1} - Y_l\|_2^2}{P(N \geq l)} = \infty$$

2. We want to show

$$\frac{\|Y_{l-1} - Y\|_2^2}{P(N \geq l)} < \infty. \quad (1)$$

By the strong order p assumption and the fact that we have a geometric time step, we have

$$\begin{aligned} \frac{\|Y_{l-1} - Y\|_2^2}{P(N \geq l)} &\leq \frac{\left(C \left(\frac{T}{2^{l-1}}\right)^p\right)^2}{q^l} \\ &= \underbrace{CT^{2p}2^{2p}}_{\tilde{C}} \left(\frac{1}{2^{2p}q}\right)^l. \end{aligned}$$

In order for this geometric series to converge, we need $\frac{1}{2^{2p}q} < 1$, i.e. $q > 2^{-2p}$.

We now procede to show that $\bar{Z}$ has finite variance. By definition, $Var[\bar{Z}] = E[\bar{Z}^2] - [E\bar{Z}]^2$. But by Jensen's inequality, we have

$$E[\bar{Z}] = E[Y] \leq (E[Y^2])^{1/2} < \infty \quad \text{because } Y \in L^2(\Omega).$$

So we just need to show $E[\bar{Z}^2] < \infty$. But

$$\begin{aligned} E[\bar{Z}^2] &= \sum_{l=0}^{\infty} \frac{\bar{v}_l}{P(N \geq l)} \\ &= \sum_{l=0}^{\infty} \frac{\|Y_{l-1} - Y\|_2^2 - \|Y_l - Y\|_2^2}{P(N \geq l)} \\ &\leq \sum_{l=0}^{\infty} \frac{\|Y_{l-1} - Y\|_2^2}{P(N \geq l)} < \infty \end{aligned}$$